

HYDROGENATED NBR (HNBR) SYNTHESIS AND LOW-MOLECULAR-WEIGHT PROCESS PATENTS

1. ABSTRACT

This report synthesizes five primary patents relevant to hydrogenated nitrile butadiene rubber (HNBR), with emphasis on routes that reduce residual unsaturation, improve heat/oil resistance, and control molecular weight and processability. The patent landscape splits into two major themes: (1) direct hydrogenation of NBR using noble-metal catalysts under controlled pressure, temperature, and catalyst loading, and (2) combined metathesis/hydrogenation strategies that reduce molecular weight during or before hydrogenation to obtain lower-Mooney HNBR. Across the set, the target material is consistently defined as an NBR-derived elastomer containing at least one conjugated diene and at least one α,β -unsaturated nitrile, with hydrogenation levels typically described as partial to near-complete saturation of the diene-derived double bonds. Several patents explicitly target low residual double bond content (RDB), narrower polydispersity, and lower Mooney viscosity than conventional HNBR grades.

US7951875B2 and EP3827029A1 focus on hydrogenation of NBR in solution using ruthenium, palladium, or rhodium catalysts, with explicit attention to catalyst residue management, aging stability, and product discoloration. EP3827029A1 provides the clearest operating window, specifying catalyst metal contents, hydrogenation temperature, pressure, and time. EP2473280B1 and US9150669B2 disclose single-step or coupled metathesis/hydrogenation processes using ruthenium/osmium catalysts, optionally with co-olefins, to simultaneously cleave unsaturation and hydrogenate the polymer, thereby lowering molecular weight and narrowing molecular weight distribution. US8957161B2 is adjacent rather than direct HNBR hydrogenation: it teaches salt-accelerated metathesis of nitrile rubber using ruthenium metathesis catalysts, which is relevant because metathesis degradation is a precursor step used in low-MW HNBR manufacture.

Overall, the evidence indicates that effective HNBR synthesis benefits from solution-phase processing, ruthenium-based catalyst systems, controlled hydrogen pressure and temperature, and deliberate management of catalyst loading and post-reaction purification. For low-MW HNBR, the strongest evidence supports combining metathesis with hydrogenation and using feedstocks with moderate Mooney viscosity to preserve processability while achieving low RDB and reduced molecular weight.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US8957161B2
- Patent Title: Process for reacting chemical compounds in the presence of catalyst systems
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses metathesis of nitrile rubber in the presence of a ruthenium- or osmium-based metathesis catalyst system and one or more salts that accelerate catalyst activity. The process is aimed at degrading nitrile rubber before or in conjunction with hydrogenation to facilitate low-molecular-weight HNBR manufacture, rather than hydrogenating NBR directly in a single step. - Salt(s):metathesis catalyst weight ratio: 0.01:1 to 10000:1 — Claim 1 - Example 7 LiBr addition: 1 mg — Example 7 - Example 7 Grubbs II catalyst: 10 mg — Example 7 - Example 7 chlorobenzene: 0.3 ml — Example 7 - Example 7 CDCl₃: 0.2 ml — Example 7 - Example 7 diethyl diallylmalonate addition: 0.15 ml — Example 7 - Example 7 reaction temperature: room temperature — Example 7 - Example 8 CsBr addition: 1 mg — Example 8 - Example 8 Grubbs II catalyst: 10 mg — Example 8 - Example 8 chlorobenzene: 0.3 ml — Example 8 - Example 8 CDCl₃: 0.2 ml — Example 8 - Example 8 diethyl diallylmalonate addition: 0.15 ml — Example 8 - Example 8 reaction temperature: room temperature — Example 8

Relevant Experimental Evidence - Example 7: LiBr accelerated ring-closing metathesis of diethyl diallylmalonate relative to the no-salt control. - Example 8: CsBr also accelerated ring-closing metathesis, but less strongly than LiBr in the reported time course.

Technical Relevance This patent is relevant because low-MW HNBR routes often rely on metathesis degradation of NBR prior to hydrogenation. It provides catalyst-activation concepts and salt effects that can be leveraged to control chain scission in HNBR precursor processing.

Patent 2

- Patent Number: US7951875B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US

- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent teaches preparation of low molecular weight hydrogenated nitrile rubber by metathesis degradation of nitrile rubber using co-olefins, followed by hydrogenation. The process is carried out in solution and emphasizes reduced molecular weight and narrower molecular weight distribution in the final HNBR product. - Rubber mass: 200 g — Example 1 - MCB: 1133 g — Example 1 - Solid content: 15 wt.-% — Example 1 - Ethylene pressure: 6.9×10^5 Pa — Example 1 - Degassing cycles: 3 — Example 1 - Rubber mass: 200 g — Example 2 - MCB: 1133 g — Example 2 - Solid content: 15 wt.-% — Example 2 - Ethylene pressure: 6.9×10^5 Pa — Example 2 - Degassing cycles: 3 — Example 2 - Rubber mass: 75 g — Example 3 - MCB: 1175 g — Example 3 - Solid content: 6 wt.-% — Example 3 - Ethylene pressure: 6.9×10^5 Pa — Example 3 - Degassing cycles: 3 — Example 3 - Rubber mass: 75 g — Example 4 - MCB: 1175 g — Example 4 - Solid content: 6 wt.-% — Example 4 - 1-hexene: 150 g — Example 4 - Degassing cycles: 3 — Example 4 - Catalyst load: 0.05 phr — Table 1 - Catalyst load: 0.10 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Reactor temperature: 80° C. — Table 1 - Co-olefin pressure: 500 psi — Table 1 - Mn: 27 kg/mol — Table 2 summary text - Mw: 54 kg/mol — Table 2 summary text - Starting polymer Mn: 85 kg/mol — Table 2 summary text - Starting polymer Mw: 296 kg/mol — Table 2 summary text - PDI: 3.4 — Table 2 summary text - PDI: 2.0 — Table 2 summary text - PDI: 1.90 — Table 2 summary text - Mooney viscosity: 73 — Table 2 summary text - Mooney viscosity: 43 — Table 2 summary text - Mooney viscosity: 28 — Table 2 summary text - Mooney viscosity: 3 — Table 2 summary text - Mooney viscosity: 31 — Table 2 summary text

Relevant Experimental Evidence - Example 1: ethylene-assisted metathesis/hydrogenation was run on a 15 wt.-% cement with 3 degassing cycles and 6.9×10^5 Pa ethylene. - Example 2: the same general conditions as Example 1 were repeated, supporting reproducibility at the same catalyst/pressure regime. - Example 3: lower-solids cement (6 wt.-%) was processed under ethylene, demonstrating strong molecular-weight reduction. - Example 4: 1-hexene was used as co-olefin at 80° C., showing that alternative chain-transfer partners also produce low-MW HNBR with reduced PDI.

Technical Relevance This patent is directly relevant because it couples metathesis and hydrogenation to generate low-MW HNBR in a single process. It provides concrete operating conditions and property outcomes showing how co-olefin choice, catalyst loading, and temperature affect molecular weight and Mooney viscosity.

Patent 3

- Patent Number: EP3827029A1
- Patent Title: Method for producing hydrogenated nitrile rubber and hnbr compositions thereof
- Assignee: Not disclosed

- Jurisdiction: EP
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses solution hydrogenation of at least partially unsaturated nitrile rubber using ruthenium, palladium, or rhodium compounds, with emphasis on minimizing residual catalyst while maintaining hydrogenation performance. The process is designed to produce HNBR with improved aging properties and reduced discoloration, while operating within defined catalyst-metal, temperature, pressure, and time windows. - Ruthenium amount: 10 ppm to 200 ppm — Claim 1 - Ruthenium amount: 10 ppm to 150 ppm — Claim 1 - Ruthenium amount: 10 ppm to 120 ppm — Claim 1 - Ruthenium amount: 10 ppm to 79 ppm — Claim 1 - Ruthenium amount: 42 ppm to 79 ppm — Claim 1 - Palladium amount: 20 ppm to <67 ppm — Claim 1 - Palladium amount: 44 ppm to <67 ppm — Claim 1 - Rhodium amount: 50 ppm to <270 ppm — Claim 1 - Rhodium amount: 79 ppm to <270 ppm — Claim 1 - Rhodium amount: 50 ppm to 250 ppm — Claim 1 - Rhodium amount: 50 ppm to 170 ppm — Claim 1 - Rhodium amount: 79 ppm to 170 ppm — Claim 1 - Hydrogenation temperature: 60 to 200°C — Claim 1 - Hydrogenation pressure: 700000 pascals to 15000000 pascals — Claim 1 - Hydrogenation time: 1 to 24 hours — Claim 1 - Diene content: at least 40% to 90% by weight — Claim 1 - Unsaturated nitrile content: at least 10% to 60% by weight — Claim 1 - Rhodium content in prior art HNBR: 58-60 ppm — Background - Removal rate: 99% — Background (CN-A-101704926 cited prior art)

Relevant Experimental Evidence - No experimental evidence extracted from available evidence.

Technical Relevance This is one of the most directly relevant patents because it explicitly covers HNBR production by hydrogenation of NBR in solution. It also provides practical catalyst-metal ranges and operating windows that are useful for balancing hydrogenation rate, catalyst residue, and product aging stability.

Patent 4

- Patent Number: EP2473280B1
- Patent Title: Process for the preparation of hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent teaches simultaneous hydrogenation and metathesis of nitrile rubber in the presence of hydrogen and optionally a co-olefin, using ruthenium- or osmium-based catalysts. The process is intended to produce HNBR with lower molecular weight, lower Mooney viscosity, and narrower molecular weight distribution in a single step. - Mw: 20,000 to 250,000 — Description - Mooney viscosity (ML 1+4 @ 100°C): 1 to 50 — Description - MWD (polydispersity index): less than 2.8 — Description - Mooney viscosity (ML 1+4 at 100°C): 24 to 70 — Description - Mooney viscosity (ML 1+4 at 100°C): 28 to 40 — Description - Catalyst formula: general formulae (I), (II) or (III) with M = ruthenium or osmium — Claim 1 - Example 1 catalyst amount: 0,057 phr — Example 1 - Example 1 catalyst amount: 0,080 phr — Example 1 - Example 1 catalyst amount: 0,06 phr — Example 1 - Example 1 catalyst amount: 0,055 phr — Example 1 - Example 1 metal content: 65,6 ppm — Example 1 - Example 1 metal content: 91,3 ppm — Example 1 - Example 1 reaction time: 0 min — Example 1 - Example 1 reaction time: 60 min — Example 1 - Example 1 reaction time: 120 min — Example 1 - Example 1 reaction time: 180 min — Example 1 - Example 1 reaction time: 240 min — Example 1 - Example 1 RDB: 100% at 0 min — Example 1 - Example 1 RDB: 2,0% at 60 min — Example 1 - Example 1 RDB: 0,4% at 120 min — Example 1 - Example 1 RDB: 0,3% at 180 min — Example 1 - Example 1 RDB: 0,6% at 240 min — Example 1 - Example 1 Mw: 191541 g/mol — Example 1 - Example 1 Mn: 68643 g/mol — Example 1 - Example 1 PDI: 2,79 — Example 1 - Example 1 Mooney viscosity: 38,2 — Example 1 - Example 1 Mw: 135772 g/mol — Example 1 - Example 1 Mn: 58259 g/mol — Example 1 - Example 1 PDI: 2,33 — Example 1 - Example 1 Mooney viscosity: 20,2 — Example 1 - Example 1 Mw: 195304 g/mol — Example 1 - Example 1 Mn: 64130 g/mol — Example 1 - Example 1 PDI: 3,05 — Example 1 - Example 1 Mooney viscosity: 69,3 — Example 1 - Example 1 Mw: 144212 g/mol — Example 1 - Example 1 Mn: 59851 g/mol — Example 1 - Example 1 PDI: 2,45 — Example 1 - Example 1 Mooney viscosity: 32,6 — Example 1

Relevant Experimental Evidence - Example 1: compared Grubbs III, Wilkinson's, and Grubbs II catalysts at similar metal contents and showed faster hydrogenation for Grubbs III in the first 120 minutes. - Example 1: demonstrated that catalyst choice strongly affects RDB decay, molecular weight, and Mooney viscosity in the final HNBR.

Technical Relevance This patent is highly relevant because it directly addresses the target transformation: simultaneous hydrogenation and metathesis of NBR to HNBR. It provides a practical route to low-RDB, lower-MW HNBR and shows how catalyst identity and loading influence final polymer properties.

Patent 5

- Patent Number: US9150669B2
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed

- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a process for producing low molecular weight HNBR by contacting nitrile rubber with a ruthenium- or osmium-containing catalyst in the presence of hydrogen and optionally a co-olefin. The process is positioned as an alternative to earlier Grubbs-type systems and is intended to improve hydrogenation efficiency while reducing molecular weight and narrowing molecular weight distribution. - Mw: 20,000 to 250,000 — Description - Mooney viscosity (ML 1+4 @ 100°C): 1 to 50 — Description - MWD (polydispersity index): less than 3.0 — Description - Catalyst formula: general formula (I) with M = ruthenium or osmium — Claim 1 - Example 1 catalyst amount: 0.041 phr — Example 1 - Example 1 catalyst amount: 0.045 phr — Example 1 - Example 1 catalyst amount: 0.060 phr — Example 1 - Example 1 metal content: 65.6 ppm — Example 1 - Example 1 co-catalyst: NIL — Example 1 - Example 1 co-catalyst amount: NIL — Example 1 - Example 1 co-catalyst: triphenylphosphine — Example 1 - Example 1 co-catalyst amount: 1.0 phr — Example 1 - Example 1 reaction time: 0 min — Example 1 - Example 1 reaction time: 60 min — Example 1 - Example 1 reaction time: 120 min — Example 1 - Example 1 reaction time: 180 min — Example 1 - Example 1 reaction time: 240 min — Example 1 - Example 1 RDB: 100% at 0 min — Example 1 - Example 1 RDB: 2.5% at 60 min — Example 1 - Example 1 RDB: 0.6% at 120 min — Example 1 - Example 1 RDB: 0.5% at 180 min — Example 1 - Example 1 RDB: 0.4% at 240 min — Example 1 - Example 1 RDB: 7.0% at 60 min — Example 1 - Example 1 RDB: 4.5% at 120 min — Example 1 - Example 1 RDB: 3.9% at 180 min — Example 1 - Example 1 RDB: 3.5% at 240 min — Example 1 - Example 1 RDB: 15.4% at 60 min — Example 1 - Example 1 RDB: 3.1% at 120 min — Example 1 - Example 1 RDB: 1.3% at 180 min — Example 1 - Example 1 RDB: 0.7% at 240 min — Example 1 - Example 1 Mw: 113371 g/mol — Example 1 - Example 1 Mn: 48352 g/mol — Example 1 - Example 1 PDI: 2.34 — Example 1 - Example 1 Mooney viscosity: 33.3 — Example 1 - Example 1 Mw: 133758 g/mol — Example 1 - Example 1 Mn: 54595 g/mol — Example 1 - Example 1 PDI: 2.45 — Example 1 - Example 1 Mooney viscosity: 41.3 — Example 1 - Example 1 Mw: 195304 g/mol — Example 1 - Example 1 Mn: 64130 g/mol — Example 1 - Example 1 PDI: 3.05 — Example 1 - Example 1 Mooney viscosity: 69.3 — Example 1

Relevant Experimental Evidence - Example 1: compared Hoveyda, Arlt, and Wilkinson's catalysts and showed that the invention catalyst systems reduced RDB efficiently while lowering molecular weight. - Example 1: demonstrated that the catalyst system with no co-catalyst outperformed the Wilkinson's system in hydrogenation efficiency and product viscosity reduction.

Technical Relevance This patent is directly relevant to low-MW HNBR synthesis because it combines hydrogenation with metathesis in one step. It provides comparative catalyst data showing how catalyst structure and co-catalyst choice affect hydrogenation rate, residual unsaturation, and final polymer molecular weight.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US8957161B2 is a metathesis-acceleration patent for nitrile rubber degradation, while US7951875B2, EP2473280B1, and US9150669B2 integrate metathesis with hydrogenation to directly generate low-MW HNBR.
- EP3827029A1 is the most conventional hydrogenation patent in the set: it uses solution hydrogenation of NBR with Ru/Pd/Rh catalysts and focuses on catalyst residue control rather than metathesis-driven chain scission.
- Catalyst families differ by function: US8957161B2 uses Ru/Os metathesis catalysts plus salts; US7951875B2 and the two US/EP low-MW HNBR patents use Ru/Os metathesis/hydrogenation catalysts; EP3827029A1 uses Ru, Pd, or Rh hydrogenation catalysts.
- Operating windows are most explicit in EP3827029A1 (60 to 200°C, 700000 to 15000000 pascals, 1 to 24 hours) and US7951875B2 (80°C, 500 psi co-olefin, 6.9×10^5 Pa ethylene, 0.05 to 0.25 phr catalyst).
- Property targets converge on low residual unsaturation and improved processability: EP2473280B1 and US9150669B2 explicitly target Mw 20,000 to 250,000 and Mooney viscosity 1 to 50, while US7951875B2 reports Mooney values as low as 3 and PDI down to 1.90.
- Co-olefin use is explicit in US7951875B2 and US9150669B2, supporting chain-transfer/metathesis control; EP2473280B1 also allows optional co-olefin, whereas EP3827029A1 does not emphasize co-olefin-assisted chain scission.
- Catalyst residue management is a major differentiator: EP3827029A1 centers on minimizing and removing noble-metal residues, while the metathesis/hydrogenation patents emphasize catalyst choice and reaction design to achieve the desired molecular-weight reduction in situ.

4. CONCLUSION

The patent evidence shows two viable HNBR synthesis strategies: direct solution hydrogenation of NBR with Ru/Pd/Rh catalysts, and single-step or staged metathesis/hydrogenation using Ru/Os catalysts to lower molecular weight while saturating residual double bonds. For conventional HNBR production, EP3827029A1 provides the clearest practical hydrogenation window and catalyst-metal ranges, with strong emphasis on minimizing residual noble metals and preserving aging performance. For low-MW HNBR, EP2473280B1 and US9150669B2 provide the strongest evidence that simultaneous metathesis and hydrogenation can deliver Mw around 20,000 to 250,000 g/mol, Mooney viscosity in the 1 to 50 range, and low PDI, especially when catalyst identity and co-olefin selection are controlled. US7951875B2 further supports the use of ethylene or 1-hexene, moderate temperature, and low phr catalyst loadings to achieve substantial molecular-weight reduction. Recommended synthesis parameters based on the evidence are solution-phase processing, Ru-based catalysts for low-MW routes, hydrogenation temperatures around 60 to 200°C, hydrogen pressures in the 700000 to 15000000 pascal range, and deliberate control of catalyst loading and co-olefin feed to balance hydrogenation rate, residual unsaturation, and polymer integrity.

5. REFERENCES

- US8957161B2 | Process for reacting chemical compounds in the presence of catalyst systems | Not disclosed | US | 2015 | <https://patents.google.com/patent/US8957161B2/en>
- US7951875B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2011 | <https://patents.google.com/patent/US7951875B2/en>
- EP3827029A1 | Method for producing hydrogenated nitrile rubber and hnbr compositions thereof | Not disclosed | EP | 2021 | <https://patents.google.com/patent/EP3827029A1/en>
- EP2473280B1 | Process for the preparation of hydrogenated nitrile rubber | Not disclosed | EP | 2014 | <https://patents.google.com/patent/EP2473280B1/en>
- US9150669B2 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2015 | <https://patents.google.com/patent/US9150669B2/en>